Database

- users: DataList<User>
- requests: DataList<ReliefRequest>
- blacklist: DataList<User>
- hurricanes: DataList<Hurricane>
- shelters: DataList<Shelter>
- contacts: DataList<Contact>
- locations: DataList<Location>
- livingBeings: DataList<LivingBeing>
- volunteers: DataList<Volunteer>
- shelterManagers: DataList<ShelterManager>
- dataLists: HashMap<Class<? extends Serializable>, DataList<? extends Serializable>>
- database: Database

- Database()
+ getInstance(): Database
+ read(): void
+ write(): void
+ write(Class<? extends Serializable> key): void
+ readDataLists(): void
+ main(String[] args): void
- resolveReferences(): void
+ attemptLogin(String email, String password): User
+ createAccount(String firstName, String lastName, LocalDate dateOfBirth, ArrayList<ReliefType> reliefNeeds, String password, Location location, Contact userContact, Contact emergencyContact): boolean
+ createReliefRequest(User author, String description, Location location, ArrayList<LivingBeing> livingBeings, Contact contactInfo): boolean
+ getUsers(Filter<User> filter): ArrayList<User>
+ getRequests(Filter<ReliefRequest> filter): ArrayList<ReliefRequest>
+ getHurricanes(Filter<Hurricane> filter): ArrayList<Hurricane>
+ getShelters(Filter<Shelter> filter): ArrayList<Shelter>
+ getContacts(Filter<Contact> filter): ArrayList<Contact>
+ getLocations(Filter<Location> filter): ArrayList<Location>
+ getLivingBeingByUUID(UUID id): LivingBeing
+ getUserByUUID(UUID id): User
+ getVolunteerByUUID(UUID id): Volunteer
+ getReliefRequestByUUID(UUID id): ReliefRequest
+ getHurricaneByUUID(UUID id): Hurricane
+ getShelterByUUID(UUID id): Shelter
+ getContactByUUID(UUID id): Contact
+ getLocationByUUID(UUID id): Location
+ getSortedReliefRequests(PriorityFilter<ReliefRequest> filter): ArrayList<ReliefRequest>
+ getSortedShelters(ReliefRequest task): ArrayList<Shelter>
- countMatchingAttributes(Shelter shelter, ReliefRequest task): int
+ exportTripInformation(Volunteer volunteer, Shelter shelter): boolean

Arrows omitted for clarity:
User
ReliefRequest
Hurricane
Shelter
Location
Contact
LivingBeing
ShelterManager
Volunteer

DataManager<T implements Serializable>

- TXT_DESTINATION: String
- parser: JSONParser
- filename: String
- mode: FileMode

+ DataManager(String filename)
+ read(): ArrayList<?>
+ write(ArrayList<?> objects): boolean
+ readJSON(): ArrayList<JSONObject>
+ writeJSON(ArrayList<JSONObject> objects): boolean
- formattedWriteJSON(String unformattedJson): String
+ readUUIDList(): ArrayList<JSONObject>
+ write(JSONObject object): void
+ writeTxt(String toWrite, String outputFileName): boolean

Serializable

+ file: String
id: UUID
serializeBehavior: SerializeBehavior

+ Serializable()
+ serialize(): Object
+ deserialize(Object object): void
+ resolveReferences(): void

SerializeBehavior<T extends Serializable>

serializable: T

+ SerializeBehavior(T serializable)
+ serialize(): Object
+ deserialize(Object object): void
+ resolveReferences(): void

SerializeJSON<T extends Serializable>

serializable: T

+ SerializeBehavior(T serializable)
+ serialize(): JSONObject
+ deserialize(JSONObject jsonObject): void
+ deserialize(Object object) void

Classes Ommitted for clarity:
SerializeJSONUser
SerializeJSONContact
SerializeJSONHurricane
SerializeJSONLivingBeing
SerializeJSONLocation
SerializeJSONReliefRequest
SerializeJSONShelter
SerializeJSONShelterManager
SerializeJSONUser
SerializeJSONVolunteer

Arrows omitted for clarity:
User
ReliefRequest
Hurricane
Shelter
Contact
LivingBeing
Location
ShelterManager
Volunteer

DataList<T extends Serializable>

- dataLists: HashMap<Class<? extends Serializable>, DataList<? extends Serializable>>
- objects: ArrayList<T>
- objectsById: HashMap<UUID, T>
- fileName: String
- dataManager: DataManager
- cls: Class<T>

- DataList<T>()
+ getInstance(String filename): DataList
+ read(): void
+ write(): void
+ resolveReferences(): void
+ filterByAttribute(Filter<T> filter): ArrayList<T>
+ getObjects(Filter<T> filter): ArrayList<T>
+ getObject(Filter<T> filter): T
+ getObjectByUUID(UUID id): T
+ add(T object): boolean
+ remove(T object): boolean
+ save(): boolean
+ exportTripInformation(String toWrite, String outputFileName): boolean

<<interface>>
Filter<T>

filter(T object): boolean

<<enumeration>>
JSONFile

BLACKLIST
CONTACTS
HURRICANES
REQUESTS
SHELTERS
USERS
VOLUNTEERS
LIVING_BEINGS
SHELTER_MANAGERS

Arrows omitted for clarity:
User
ReliefRequest
Hurricane
Shelter
Contact
LivingBeing
Location
ShelterManager
Volunteer

ConsoleUI

- app: Application
- scanner: Scanner

+ ConsoleUI()
+ run(): void
- pauseForNextScenario(String nextScenarioName): void
+ close(): void
- printHeader(String title): void
- printResult(boolean success, String successMessage, String failureMessage): void
+ loginScenario(String email, String password): boolean
- printReliefRequests(): void
+ readAndPrintSheltersScenario(): void
+ createReliefRequestScenario(): void
+ openAndMatchRequestScenario(): void
+ findShelterAndAssignRequest(): void
+ exportTripInformationScenario(): void
+ rolePermissionScenario(): void
+ checkIntoShelterScenario(): void
+ createAccountAndLoginScenario(): void
+ getShelterTimes(): void
+ main(String[] args): void

Application

- application: Application
- database: Database
- currentUser: User
- location: Location
- currShelter: Shelter
- pendingGroupMembers: ArrayList<LivingBeing>

- Application()
+ getInstance(): Application
+ requestCreateAccount(String firstName, String lastName, String dateOfBirth, String password, String address, String userContactEmail, String userContactPhone, String userContactPreference, String emergencyContactEmail, String emergencyContactPhone, String emergencyContactPreference): boolean
+ attemptLogin(String email, String password): String
+ getUserString(): String
+ getVolunteerProfile(): String
+ logout(): void
+ getUserFullName(): String
+ createReliefRequest(String description): boolean
- openReliefRequest(ReliefRequest reliefRequest): boolean
+ openReliefRequest(int index): boolean
+ acceptCurrentReliefRequest(): boolean
+ acceptReliefRequest(int index): boolean
- acceptReliefRequest(ReliefRequest reliefRequest): boolean
+ closeReliefRequest(): boolean
+ getShelters(): ArrayList<String>
+ getRequests(): ArrayList<String>
+ getHurricaneData(): ArrayList<String>
+ getContacts(): ArrayList<String>
+ getShelterTimes(): ArrayList<Double>
+ checkIntoShelter(): boolean
+ getCurrentReliefRequestString(): String
- getCurrentReliefRequest(): ReliefRequest
+ getReliefRequestStrings(): ArrayList<String>
- getSortedShelters(): ArrayList<Shelter>
+ getSortedShelterStrings(): ArrayList<String>
+ selectShelter(int index): boolean
+ exportTripInformation(): boolean
+ startNewReliefGroup(): void
+ addMemberToReliefGroup(String firstName, String lastName, String dateOfBirth, String type, ArrayList<String> reliefNeeds): void
+ submitGroupReliefRequest(String description): boolean

Arrows omitted for clarity:
User
LivingBeing
Location
Shelter

Shelter

+ file: String
- name: String
- location: Location
- contactInfo: Contact
- capacity: int
- occupants: ArrayList<LivingBeing>
- occupantsByUUID: ArrayList<UUID>
- volunteers: ArrayList
- volunteersByUUID: ArrayList<UUID>
- status: ShelterStatus
- isPetFriendly: boolean
- manager: ShelterManager
- locationUUIDReference: UUID
- managerUUIDReference: UUID
- contactUUIDReference: UUID
- timeTo: double

+Shelter(UUID id, String name, Location location, int capacity, ArrayList volunteers, ShelterStatus status, Contact contactInfo, boolean isPetFriendly)
+ Shelter(String name, Location location, int capacity, ArrayList volunteers, ShelterStatus status, Contact contactInfo,boolean isPetFriendly)
+ Shelter()
+ getNumOccupants(): int
+ hasCapacity(int partySize): boolean
+ addOccupant(LivingBeing occupant): boolean
+ removeOccupant(LivingBeing occupant): boolean
+ addVolunteer(Volunteer volunteer): void
+ removeVolunteer(Volunteer volunteer): void
+ checkIn(ArrayList livingBeings): boolean
+ hasMedicalStaff(): boolean

<<enumeration>>
ShelterStatus

OPEN
CLOSED
FULL
TEMP_UNAVAILABLE

<<enumeration>>
HurricaneStatus

APPROACHING
LANDFALL
PASSED

Hurricane

+ file: String
- name: String
- category: int
- locations: ArrayList<Location>
- locationsUUIDReference: ArrayList<Location>
- status: HurricaneStatus

+ Hurricane()
+ Hurricane(String name, int category, ArrayList locations, HurricaneStatus status)
+ Hurricane(UUID id, String name, int category, LocalDateTime startDate, LocalDateTime endDate, ArrayList locations, HurricaneStatus status)
+ addLocation(Location location): void
+ removeLocation(Location location): void

Location

+ file: String
- state: String
- city: String
- street: String
- zip: String

+ Location(String state, String city, String street, String zipcode)
+isInState(String state): boolean
+isInCity(String city): boolean
+ isInZip(String zipcode): boolean
+ getCoordinates(): Coordinates

Contact

- file: String
- email: String
- phone: String
- contactPreference: ContactPreference

+ Contact(UUID id, String email, String phoneNumber, ContactPreference contactPreference)
+ Contact(String email, String phoneNumber, ContactPreference contactPreference)
+ Contact()

<<enumeration>>
ContactPreference

TEXT
CALL
EMAIL

<<enumeration>>
LivingBeingType

HUMAN,
PET

LivingBeing

+ file: String
- firstName: String
- lastName: String
- dateOfBirth: LocalDate
- type: LivingBeingType
- reliefNeeds: ArrayList<ReliefType>

+ LivingBeing(String firstName, String lastName, LocalDate dateOfBirth, LivingBeingType type, ArrayList reliefNeeds)
+ LivingBeing()
+ addReliefNeed(ReliefType need): boolean
+ removeReliefNeed(ReliefType need): boolean
+ getAge(): int

User

- password: String
- location: Location
- locationUUIDReference: UUID
- userContact: Contact
- emergencyContact: Contact
- userReliefRequests: ArrayList<ReliefRequest>
- userReliefRequsts: ArrayList<UUID>

+ User(String firstName, String lastName, LocalDate dateOfBirth, ArrayList reliefNeeds, String password, Location location, Contact userContact, Contact emergencyContact)
+ User(UUID id, String firstName, String lastName, LocalDate dateOfBirth, ArrayList reliefNeeds, String password,Location location, Contact userContact, Contact emergencyContact)
+ User()
+ addUserReliefRequest(ReliefRequest reliefRequest): boolean
+ addUserReliefRequest(UUID id): boolean
+ getFullName(): String

Volunteer

+ file: String
- isAvailable: boolean
- reliefType: ArrayList<ReliefType>
- task: ReliefRequest
- sorter: SkillSort
- taskUUIDReference: UUID

+ Volunteer(String firstName, String lastName, LocalDate dateOfBirth, ArrayList reliefNeeds, String password, Location location, Contact userContact, Contact emergencyContact, boolean isAvailable, ArrayList reliefTypes, ReliefRequest task)
+ Volunteer(UUID id, String firstName, String lastName, LocalDate dateOfBirth, ArrayList reliefNeeds, String password,Location location, Contact userContact, Contact emergencyContact, boolean isAvailable, ArrayList reliefTypes,ReliefRequest task)
+ Volunteer()
+ addReliefType(ReliefType reliefType): void
+ removeReliefType(ReliefType reliefType): void
+ acceptReliefRequest(ReliefRequest reliefRequest): boolean
+ openReliefRequest(ReliefRequest reliefRequest): boolean

ShelterManager

+ file: String
- shelter: Shelter
- shelterUUIDReference: UUID

+ ShelterManager(UUID id, String firstName, String lastName, LocalDate dateOfBirth, String password, Location location,Contact userContact, Contact emergencyContact, boolean isAvailable, ArrayList reliefTypes, ReliefRequest task)
+ ShelterManager(Volunteer volunteer)
+ ShelterManager()
+ closeReliefRequest(ReliefRequest reliefRequest): boolean

ReliefRequest

-description: String
-location: Location
-types: ArrayList<ReliefType>
-status: ReliefStatus
-contactInfo: Contact
- livingBeings: ArrayList<LivingBeing>
- contactUUIDReference: UUID
- locationUUIDReference: UUID
- assignedTo: Volunteer
- assignedToUUIDReference: UUID

+ ReliefRequest(String description, Location location, ReliefStatus status, ArrayList livingBeings, ArrayList reliefTypes,Contact contactInfo)
+ ReliefRequest(UUID id, String description, Location location, ReliefStatus status, ArrayList livingBeings, ArrayList reliefTypes, Contact contactInfo)
+ ReliefRequest()
+ addLivingBeing(LivingBeing livingBeing): boolean
+ hasPets(): boolean
+ getPartySize(): int
+ getPartySizeHumans(): int

<<enumeration>>
ReliefType

FOOD
WATER
MEDICAL_CARE
SHELTER
EVACUATION
TRANSPORTATION
OTHER

<<enumeration>>
ReliefStatus

REQUESTED
ASSIGNED
COMPLETED

<<interface>>
PriorityFilter<T>

filter(T object): boolean

SkillSort

- volunteer: Volunteer

+ SkillSort(Volunteer volunteer)
+ filter(ReliefRequest object): double